Transformations of quadratics

1 Consider the function $y = (x - 6)^2$.

a Complete the table of values.

x	4	5	6	7	8
y					

b Find the minimum value of y.

c Find the x-value that corresponds to the minimum y-value.

d Hence, state the coordinates of the vertex of the parabola.

e How many units to the right has $y = x^2$ been translated?

f Translate the parabola $y = x^2$ to sketch the graph of $y = (x - 6)^2$.

2 Consider the function $y = (x + 6)^2$.

a Complete the table of values.

x	-8	-7	-6	-5	-4
y					

b Find the minimum value of y.

c Find the x-value that corresponds to the minimum y-value.

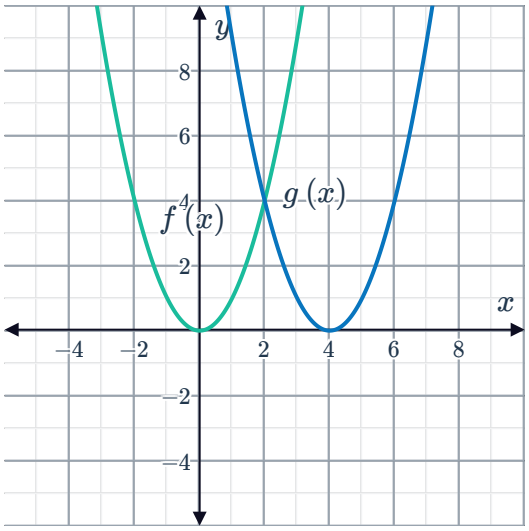
d Hence, state the coordinates of the vertex of the parabola.

e How many units to the left has $y = x^2$ been translated?

f Translate the parabola $y = x^2$ to sketch the graph of $y = (x + 6)^2$.

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3 The functions $f(x)$ and $g(x) = f(x+k)$ have been graphed on the coordinate plane.



a Determine the value of k.

b Explain what k represents on this context.

4 Consider the parabola $y = x^2 - 3$.

a Complete the table of values.

x	-2	-1	0	1	2
y					

b Use the graph of $y = x^2$ to sketch the graph of $y = x^2 - 3$.

c Find the y-intercept of the graph.

d Adding a constant to the equation $y = x^2$ corresponds to what type of transformation of its graph?

5 Consider the following quadratic functions.

i Sketch the graph of the parabola given by $h(x)$ on a coordinate plane.

ii Plot the axis of symmetry of the parabola on the same coordinate plane on part (a).

iii State the vertex of the parabola.

a $h(x) = x^2 - 2$

b $h(x) = x^2 + 2$

6

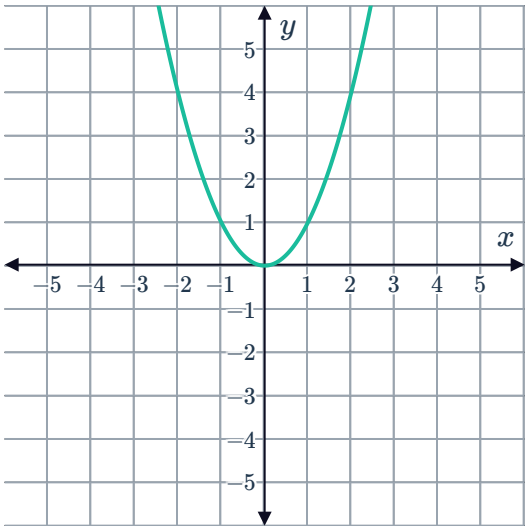
a Find the y-intercept of the parabola $y = x^2 + 3$.

b Find the y-intercept of the parabola $y = x^2 - 4$.

c How many units must the graph $y = x^2 + 3$ be shifted down to overlap the graph of $y = x^2 - 4$?

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7 Consider the given graph of $y = x^2$.



a

i How do we shift the graph of $y = x^2$ to get the graph of $y = x^2 - 3$?

ii Hence plot $y = x^2 - 3$ on the same graph as $y = x^2$.

b

i How do we shift the graph of $y = x^2$ to get the graph of $y = (x - 2)^2$?

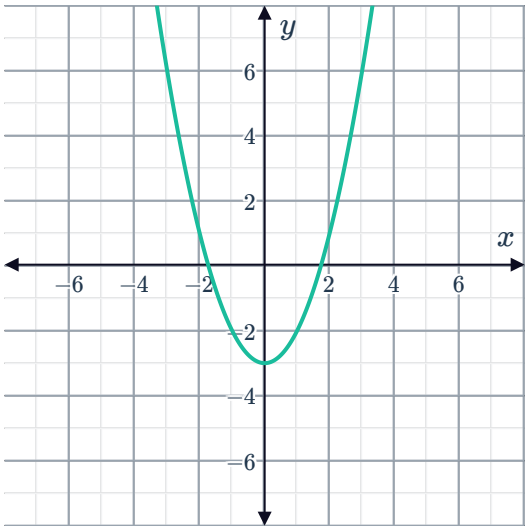
ii Hence plot $y = (x - 2)^2$ on the same graph as $y = x^2$.

8

a Find the equation of the new parabola when $y = x^2$ is translated 6 units to the right and 2 units down.

b Sketch the graph of the parabola formed after the translations.

9 Consider the graph of $y = x^2 - 3$.



a Determine the new equation when the graph of y is moved upwards by 5 units.

b Determine the new equation when the graph of y is moved to the right by 3 units.

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10 Consider the function $y = -(x-2)^2$.

a Complete the table of values.

x	0	1	2	3	4
y					

b Find the maximum value of y.

c Find the x-value that corresponds to the maximum y-value.

d Hence, state the vertex of the parabola.

e Translate the parabola $y = x^2$ to sketch $y = -(x-2)^2$.

11 Consider the parabola $y = -x^2 + 2$.

a Sketch the graph of the parabola.

b State the vertex of the parabola.

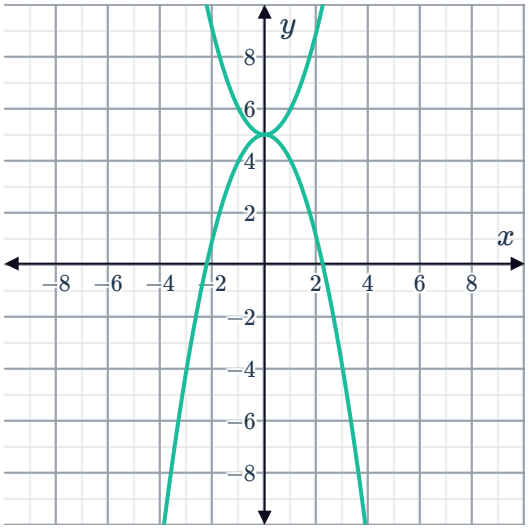
12 Consider the two graphs on the same coordinate plane.
One of them has equation $y = x^2 + 5$.
Determine whether the following could be the equation of the other graph.

a $y = \dfrac{x^2}{2} + 5$

b $y = x^2 - 5$

c $y = -\dfrac{x^2}{2} - 5$

d $y = -x^2 + 5$



13 Consider the quadratic function $y = 16 - x^2$.

a Find the x-intercepts.

b Find the y-intercept.

c Sketch the graph of the function.

d Find the vertex of the parabola.

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14 Consider the function $y = 2x^2$.

a Complete the table of values.

x	-2	-1	0	1	2
y					

b Sketch the graph on a coordinate plane.

c For $y = ax^2$, as a increases, describe how it changes the graph of $y = x^2$.

15 A parabola of the form $y = ax^2$ goes through $(2, -4)$.

a Determine the value of a.

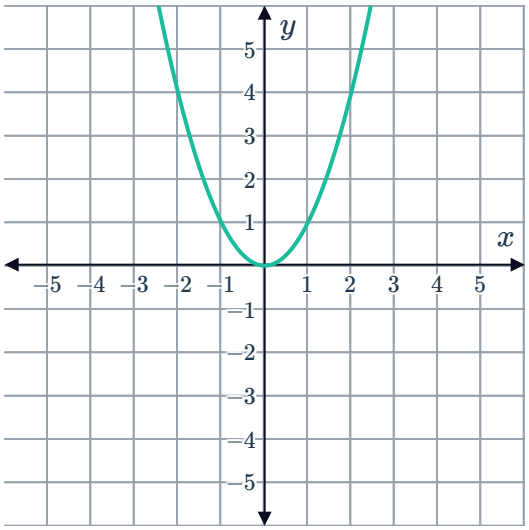
b Find the coordinates of the vertex.

c Sketch the graph of the parabola.

16 Consider the graph of $y = x^2$.

a Sketch the graph of the parabola that has the same vertex as $y = x^2$ and that passes through $(2, 2)$.

b Find the equation of the new parabola.



17 Sketch the graph of the following functions:

a $y = 4x^2 - 2$

b $y = 2(x + 4)^2$

18 Determine whether there has been a reflection on the following transformations.

a $y = x^2$ has been transformed into $y = 5x^2 + 2$.

b $y = x^2$ has been transformed into $y = 3(x - 4)^2$.

c $y = x^2$ has been transformed into $y = 5(x - 1)^2 - 3$.

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19 Identify the vertical translation on the following transformations.

a $y = x^2$ has been transformed into $y = -2x^2 + 6$

b $y = x^2$ has been transformed into $y = -4\left(x + 10\right)^2$.

c $y = x^2$ has been transformed into $y = 10\left(x - 9\right)^2 - 7$.

20 Identify the horizontal translation on the following transformations.

a $y = -5x^2$ has been transformed into $y = -5\left(x - 3\right)^2$.

b $y = x^2$ has been transformed into $y = 2\left(x + 7\right)^2 + 2$.

21 For each of the following graphs that has been transformed from $y = x^2$:

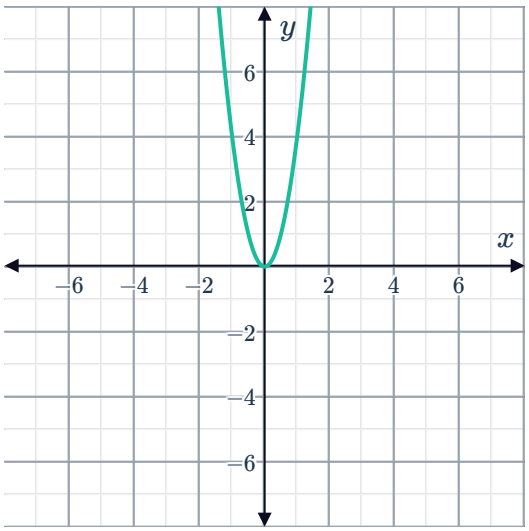
i Determine the vertical translation.

ii Determine the horizontal translation.

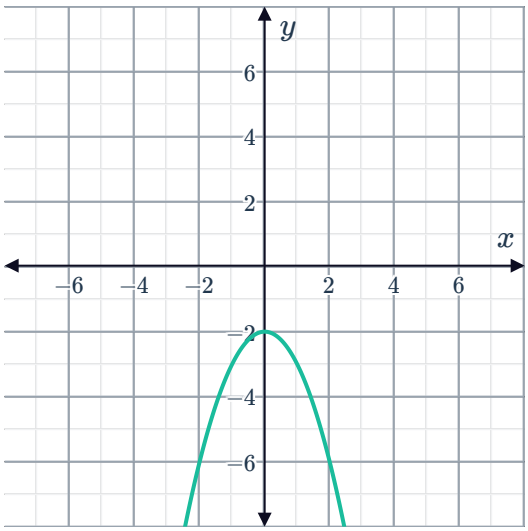
iii Has there been a reflection about the x-axis?

iv Has the quadratic had a change in dilation from the standard $y = x^2$ curve?

a

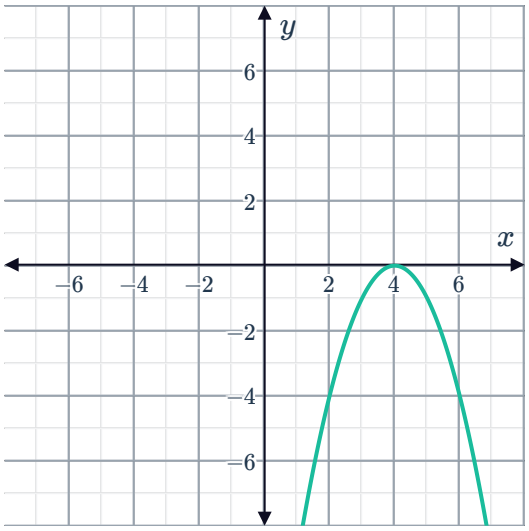


b

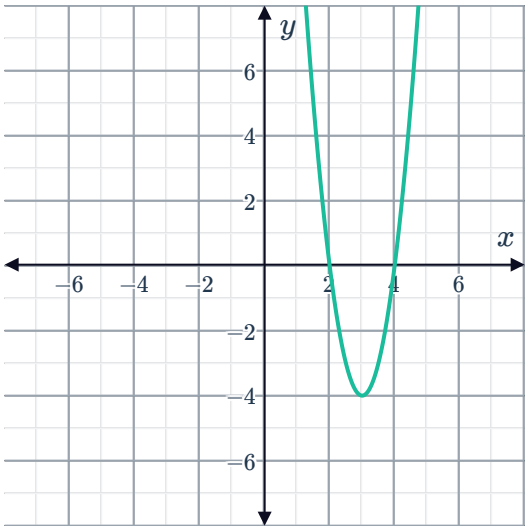


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c



d



22 Find the resulting equation when $y = x^2$ undergoes the following transformations.

a A change in dilation of 10

b A change in dilation of 5 and is reflected about the x-axis

c A vertical translation of 10 units up

d A horizontal translation of 2 units right

e A reflection about the x-axis and a vertical translation of 7 units down

f A horizontal translation of 4 units left and a change in dilation of 8

g Translated horizontally 6 units to the right and translated vertically 5 units upwards

23 Ned graphed the function $f(x) = -4(x-1)^2 + 2$ and then graphed the function $g(x) = -4(x-1)^2 - 7$ on the same coordinate plane.
Determine whether the following describes the vertex of $g(x)$.

a 9 units to the right of the vertex of $f(x)$.

b 9 units to the left of the vertex of $f(x)$.

c 9 units above the vertex of $f(x)$.

d 9 units below the vertex of $f(x)$.

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